

Alfie McGlennon

Reading | alfie.mcglennon@outlook.com

Profile

I am currently pursuing a Master's in Climate Change and Artificial Intelligence at the University of Reading, having previously completed my BSc in Meteorology and Climate at the same institution. My work integrates various disciplines, including climate science, data analysis, AI methodologies, and interactive visualization. I am particularly passionate about harnessing data-driven tools and designing accessible solutions that effectively communicate complex climate patterns in ways that are understandable and meaningful to diverse audiences.

Technical Skills

Programming & Data Tools

Python (pandas, NumPy, matplotlib), SQL, DAX, Excel, Power BI.

Data Science & Analysis

Data cleaning and preprocessing, exploratory analysis, machine learning fundamentals, anomaly detection, time series modelling.

Visualisation & Communication

Interactive dashboards (Power BI), data-driven storytelling, report design, and transparent and accessible presentation of complex findings.

Climate & Environmental Data

Experience handling gridded datasets (e.g. ERA5, Berkeley Earth), climate anomaly calculations, and interpretation of large-scale environmental trends.

Tools & Professional Skills

JupyterLab, Git, MS Office; strong communication, teamwork, and time management; confident in presenting technical material to varied audiences.

Education

University of Reading - Reading, UK | MSc

Climate Change and Artificial Intelligence, 2026

Modules include *Applied Data Science with Python*, *Artificial Intelligence and Machine Learning*, and *Data Science and Climate Services*, which combine advanced analytics and AI techniques with an understanding of the climate system.

University of Reading - Reading, UK | BSc

Meteorology and Climate, 2025

- Dissertation: Investigating an Equilibrium Approximation for the Bowen Ratio (First Class)
- Key Modules: Data Science and Climate Services, Numerical Modelling, Boundary Layer Meteorology
- Grade: 2:1

The Forest School - Wokingham | A-Levels

Mathematics (A), Geography (A), Physics (B), 2021

The Forest School - Wokingham | GCSEs

2019

I passed 11 GCSE subjects, including Mathematics (8), English Language (8), and Physics (8).

Experience

SWIM TEACHER (LEVEL 2) | 11/2020 - Current

Places Leisure - Reading

- Delivered structured instruction to students of all levels, adapting methods to diverse learning styles.
- Developed clear communication, patience and management skills.

LIFEGUARD | 08/2019 - Current

Places Leisure - Reading

- Worked as part of a team to maintain safety standards, respond to incidents, and support an inclusive environment.

DATA ANALYST INTERN | 07/2024 - 07/2024

Safestore - Borehamwood

- Analysed customer data to identify customer behaviour trends and retention metrics.
- Delivered concise presentations to senior management using PowerPoint.
- Used Python (pandas, matplotlib, seaborn) to create data visualisations to support business retention strategies.

Projects

Rainfall Extremes Dashboard (Power BI) · 2025

Independent project – Power BI, DAX, Python (pandas, matplotlib), ERA5 data

- Built an analytical dashboard exploring rainfall variability and extreme event frequency across regions using Power BI and DAX for dynamic metrics and filtering.
- Processed multi-year ERA5 precipitation data in Python to calculate anomalies, derive regional aggregates, and visualise temporal and spatial patterns.
- Created a clear, interactive interface to communicate hydroclimatic trends and extremes in a way accessible to non-technical users.

City Climate Bars & Stripes (Web App) · 2025

Independent project – Python (data pre-processing), D3.js, JavaScript, Berkeley Earth data

- Focused on large-scale data preparation and transformation of Berkeley Earth gridded temperature data for use in an AI-assisted frontend built with D3.js.
- Structured and optimised multi-decadal CSV datasets to support efficient rendering of interactive city-level "climate stripes" visualisations.
- Contributed to refining the visual and functional output, ensuring data accuracy and consistent performance across seasonal and annual modes.

Volunteering

- Coach, TVTristars Triathlon Club, 2018-2021
- Swim Instructor Support, Bulmershe Leisure Centre and Reading Swim Club, 2016-2019

Hobbies and Interests

- Triathlon competitor (10+ years, national level)
- Environmental communication
- Data storytelling
- Mentoring in sport and STEM